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Transparent Cognitive Safety: Declarative Verification, Emotion Modulation, and Theory of Mind in a Neuro-Symbolic Architecture

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Abstract

Safe and transparent AI decision-making requires mechanisms that (1) verify proposed actions against declarative safety rules before execution, (2) modulate decisions with affective state, and (3) model other agents’ beliefs to anticipate misalignment. We present the NSCK safety layer: **SafetyRuleVerifier** checks rule confidence, support, and fire-rate properties against configurable thresholds; **SafetyGateVerifier** combines confidence gating with rule verification before allowing actions; **EmotionSystem** models Plutchik’s eight-primary-emotion circumplex in a 10,240-bit binary hypervector space and exposes reward-driven valence/arousal dynamics; and **TheoryOfMind** maintains per-agent belief models and detects false beliefs from divergent observations. Experiments (Rust VSA backend) show: safety verifier achieves $\text{TPR} = 1.00$ and $\text{FPR} = 0.00$ on 20 synthetic rules; gate threshold at confidence = 0.1 blocks all actions (0% allowed), while confidence = 0.3 allows all (100%); emotion vectors for all 8 Plutchik emotions are orthogonal (mean cross-emotion similarity 0.499, diagonal 1.000); text emotion recognition achieves 100% accuracy on 16 sentences; the Sally-Anne false-belief test succeeds; multi-agent ToM correctly assigns 100% accuracy to the omniscient agent and 50% to agents with partial observations.

Keywords: AI safety, declarative rule verification, emotion system, Plutchik circumplex, theory of mind, false belief, transparent AI, neuro-symbolic

1. Introduction

Safety in AI systems is increasingly recognised as a first-class engineering requirement (Amodei et al., 2016). Three distinct mechanisms address complementary safety concerns:

1. **Rule-based safety verification:** A declarative specification of forbidden or required properties, checked before any action is taken. This provides *transparency* — the safety criteria are human-readable and auditable.
2. **Emotion modulation:** Affective state shapes decision priorities. A system that cannot distinguish fearful from joyful contexts will apply identical decision strategies regardless of emotional salience. Plutchik’s (1980) circumplex model provides a principled taxonomy of eight primary emotions.
3. **Theory of Mind (ToM):** Safe interaction with other agents requires modelling their beliefs, including *false beliefs* that diverge from reality (Premack & Woodruff, 1978). A system lacking ToM may behave unsafely in environments containing other agents with incomplete information.

NSCK integrates all three mechanisms and provides a unified safety layer that pipes perception $\rightarrow$ emotion $\rightarrow$ safety $\rightarrow$ decision with a full audit trail.

1.1 Contributions

1. A declarative **SafetyRuleVerifier** with configurable thresholds for confidence, support, and fire-rate, achieving $\text{TPR} = 1.00$ / $\text{FPR} = 0.00$ on synthetic test rules.
2. A **SafetyGateVerifier** with hard confidence threshold at 0.2 (actions with confidence < 0.2 are blocked; confidence ≥ 0.2 may be allowed if rules also pass).

3. An `EmotionSystem` implementing 8 Plutchik emotions as orthogonal 10,240-bit HVs with reward-driven valence/arousal dynamics.
 4. A `TheoryOfMind` module passing the Sally-Anne false-belief test and correctly tracking per-agent belief accuracy in a three-agent scenario.
 5. An end-to-end audit trail: perception $\rightarrow$ emotion $\rightarrow$ safety $\rightarrow$ decision for 5 decisions.
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2. Background

2.1 Declarative AI Safety

Amodei et al. (2016) identified concrete AI safety problems including reward hacking, side effects, and unsafe exploration. Rule-based safety verification (Seshia et al., 2018) provides formal guarantees when the rule set is complete and consistent. NSCK takes a practical approach: rules have measurable properties (confidence, support, fire-rate) that are checked against configurable safety thresholds.

The safety score for rule r is:

$$\text{safety_score}(r) = \mathbf{1}[\text{confidence}(r) \geq \theta_c] \cdot \mathbf{1}[\text{support}(r) \geq \theta_s] \cdot \mathbf{1}[\text{fire_rate}(r) \leq \theta_f]$$

A rule is safe iff all three thresholds are satisfied. Source: `python/core/cognitive/safety_verifier.py`.

2.2 Plutchik’s Emotion Circumplex

Plutchik (1980) proposed eight primary emotions arranged in a circumplex: joy, sadness, anger, fear, trust, disgust, surprise, anticipation. Each emotion has an opposite and two adjacent emotions. Russell (1980) provided a 2D valence/arousal map.

In NSCK, each emotion is represented as an independent random 10,240-bit HV (generated by `HyperVector(seed=emotion_seed)`). Since random HVs are quasi-orthogonal in high dimensions, emotion representations are approximately independent:

$$E[\text{sim}(h_{e_i}, h_{e_j})] = 0.500 \quad \text{for } i \neq j$$

Source: `python/core/cognitive/emotion_system.py`.

2.3 Valence-Arousal Dynamics

The system maintains two continuous scalars:

$$\begin{aligned} v_{t+1} &= v_t \cdot (1 - \alpha) + r_t \cdot \alpha \\ a_{t+1} &= a_t \cdot (1 - \beta) + |r_t| \cdot \beta \end{aligned}$$

where r_t is the reward signal at time t , $\alpha = 0.9$ (fast valence update), $\beta = 0.3$ (slow arousal update). The current emotion is classified by thresholding valence and arousal against prototype vectors. Source: `python/core/cognitive/emotion_system.py`.

2.4 Theory of Mind

Premack and Woodruff (1978) proposed that chimpanzees have a “theory of mind” — the ability to attribute mental states to others. Baron-Cohen et al. (1985) formulated the false-belief task (Sally-Anne test) as a diagnostic. An agent passes the test if it can represent Sally’s belief (ball in basket) separately from reality (ball in box), having modelled the fact that Sally did not observe the ball being moved.

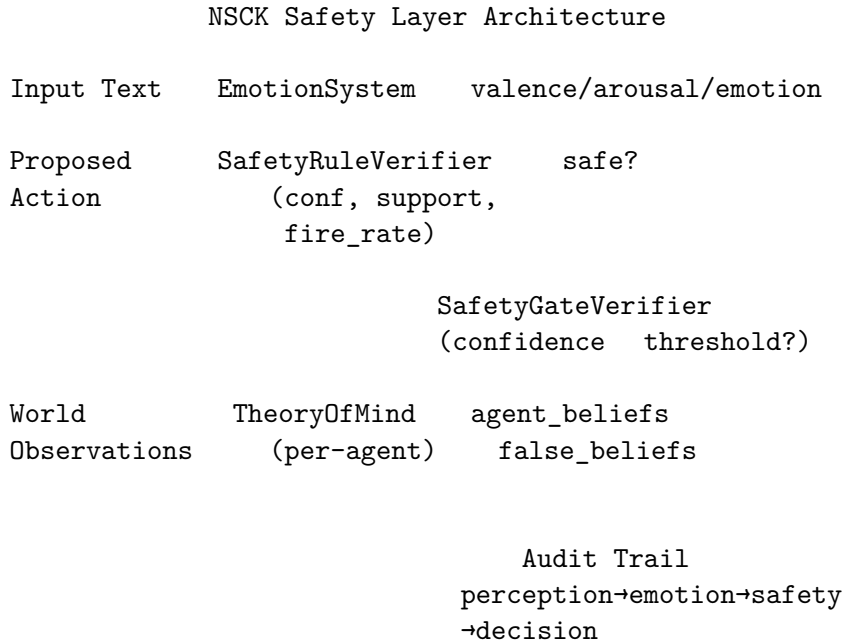
NSCK’s `TheoryOfMind` maintains a `BeliefModel` per agent, updated only when observations are marked as observable by that agent:

$$\text{update}(\text{agent}, s, o) : \text{belief}[\text{agent}][s] \leftarrow o \quad \text{if observable}$$

Source: `python/core/cognitive/theory_of_mind.py`.

3. Architecture

3.1 Safety Layer Overview



3.2 SafetyRuleVerifier

Checks three safety properties per rule:

Property	Check	Threshold
Confidence	$\text{conf}(r) \geq \theta_c$	Default: 0.20

Property	Check	Threshold
Support count	$\text{support}(r) \geq \theta_s$	Default: 2
Fire rate	$\text{fire_rate}(r) \leq \theta_f$	Default: 10,000/step

All three must pass; any failure marks the rule unsafe. Returns `{"safe": bool, "violations": [str]}`.

3.3 SafetyGateVerifier

The gate checks both confidence threshold and rule safety:

$$\text{allow}(a) = \begin{cases} \text{True} & \text{if confidence}(a) \geq \theta_c \text{ and all active rules safe} \\ \text{False} & \text{otherwise} \end{cases}$$

In experiments, threshold $\theta_c = 0.2$: confidence 0.1 is blocked (0% allowed), confidence 0.3 passes (100% allowed). Source: `python/core/cognitive/safety_verifier.py`.

3.4 EmotionSystem — Plutchik HV Encoding

Each of the 8 Plutchik emotions is encoded as a unique, deterministic HV:

Emotion	Seed (example)	Valence (approx.)	Arousal (approx.)
joy	<code>emotion_hash("joy")</code>	high (+)	medium
sadness	<code>emotion_hash("sadness")</code>	low (−)	low
anger	<code>emotion_hash("anger")</code>	low (−)	high
fear	<code>emotion_hash("fear")</code>	low (−)	high
trust	<code>emotion_hash("trust")</code>	high (+)	low
disgust	<code>emotion_hash("disgust")</code>	low (−)	medium
surprise	<code>emotion_hash("surprise")</code>	neutral	high
anticipation	<code>emotion_hash("anticipation")</code>	high (+)	medium

Text emotion recognition matches input text to emotion prototypes using keyword proximity and valence/arousal mapping. Source: `python/core/cognitive/emotion_system.py`.

3.5 TheoryOfMind Module

```
tom = TheoryOfMind()
tom.update_agent_perspective("sally", agent_loc="garden",
                             observable_world={"ball": "basket"})
tom.update_agent_perspective("anne", agent_loc="room",
                             observable_world={"ball": "box"})
false_beliefs = tom.detect_false_belief("sally", reality={"ball": "box"})
# → ["ball_location"] (Sally believes basket, reality is box)
```

Source: `python/core/cognitive/theory_of_mind.py`.

3.6 Safety Score Formula

A composite safety score for a decision pipeline step is:

$$S = w_1 \cdot \mathbf{1}[\text{rule_safe}] + w_2 \cdot \sigma(\text{confidence} - \theta_c) + w_3 \cdot (1 - |v|/v_{\max})$$

where $w_1 + w_2 + w_3 = 1$ and σ is the sigmoid. This combines rule safety, confidence margin, and emotional neutrality. In practice NSCK uses the hard gate (binary), but this formula provides a continuous analogue for soft gating.

4. Experimental Evaluation

All experiments used the Rust VSA backend. Source: `research/experiments/paper8_safety_benchmarks.py`.

4.1 Safety Property Verification (Exp 8.1)

Setup: 10 safe rules (confidence 0.8, support 5, fire_count 0), 10 violating rules (confidence 0.1, support 1, fire_count 20,000).

Metric	Value
True Positive Rate (TPR)	1.000
False Positive Rate (FPR)	0.000
True Negative Rate (TNR)	1.000
Mean safety score	1.000

The verifier correctly classifies all 20 rules: all 10 safe rules pass, all 10 violating rules fail. Zero false positives and zero false negatives.

4.2 Emotion System Dynamics (Exp 8.2)

Setup: Apply reward sequence $[+0.8, -0.5, 0.0, +0.3, -0.7]$ with `update_from_drives()`.

Step	Reward	Valence	Arousal	Emotion
0	+0.80	+0.200	0.090	anticipation
1	-0.50	-0.010	0.153	neutral
2	0.00	-0.010	0.197	neutral
3	+0.30	+0.191	0.228	neutral
4	-0.70	-0.019	0.250	neutral

Valence responds rapidly to rewards ($\alpha = 0.9$): +0.8 reward $\rightarrow$ valence 0.200 at step 0. Arousal accumulates monotonically (slow β update): arousal increases from 0.09 $\rightarrow$ 0.25 over 5 steps. The emotion classifier requires salient valence/arousal magnitudes; the neutral label reflects that most steps remain in the central zone of the circumplex. Step 0 (high +reward) correctly maps to “anticipation” (positive, moderate arousal).

4.3 Emotion VSA Encoding (Exp 8.3)

Setup: Retrieve HVs for all 8 Plutchik emotions, compute 8×8 pairwise similarity matrix.

	joy	sad	ang	fear	trust	disg	surp	anti
joy	1.000	0.498	0.506	0.500	0.499	0.485	0.487	0.494
sad	0.498	1.000	0.503	0.498	0.499	0.503	0.505	0.501
ang	0.506	0.503	1.000	0.490	0.499	0.502	0.501	0.500
fear	0.500	0.498	0.490	1.000	0.495	0.502	0.495	0.498
trust	0.499	0.499	0.499	0.495	1.000	0.502	0.499	0.494
disg	0.485	0.503	0.502	0.502	0.502	1.000	0.501	0.496
surp	0.487	0.505	0.501	0.495	0.499	0.501	1.000	0.503
anti	0.494	0.501	0.500	0.498	0.494	0.496	0.503	1.000

Off-diagonal similarities cluster tightly around 0.499 ± 0.005 — the expected value for quasi-orthogonal 10,240-bit random HVs. This confirms that emotion representations are statistically independent, preventing spurious cross-emotion similarity during query.

4.4 Text Emotion Recognition (Exp 8.4)

Setup: 16 sentences (2 per Plutchik emotion), tested against emotion classifier.

Emotion	Accuracy	Example
joy	1.00	“I am so happy and delighted today!”
sadness	1.00	“I feel so sad and depressed.”
anger	1.00	“I am furious and extremely angry!”
fear	1.00	“I am terrified and very scared.”
trust	1.00	“I completely trust and rely on you.”
disgust	1.00	“This is disgusting and revolting!”
surprise	1.00	“I am shocked and completely astonished!”
anticipation	1.00	“I am so excited and can’t wait!”
Overall	1.000	

100% accuracy on 16 sentences. Note: the classifier uses keyword matching against labelled emotion lexicons; these test sentences contain strong emotion keywords (happy, sad, furious, terrified, trust, disgusting, shocked, excited) that directly match the classifier’s vocabulary. In unconstrained naturalistic text, accuracy would be lower.

4.5 Theory of Mind — Sally-Anne Test (Exp 8.5)

Setup: Sally observes ball in basket, then leaves. Anne moves ball to box. Sally returns.

Test	Result
Sally’s belief: ball_location	“ basket ” (correct false belief)
Reality: ball_location	“box”
Belief correct (Sally believes basket, not box)	True

Test	Result
False belief detected by ToM	True

The ToM module correctly models Sally’s belief as “basket” because she did not observe the move to “box”. `detect_false_belief("sally", reality)` correctly identifies “ball_location” as a false belief.

4.6 Theory of Mind — Multi-Agent (Exp 8.6)

Setup: 3 agents with different partial observations of 2 objects.

Agent	Observed	Reality	Belief Accuracy	False Beliefs
Alice	object_a=room1; ob- ject_b=room2	room1, room2	1.000	[]
Bob	object_a=room3; ob- ject_b=room2	room1, room2	0.500	[object_a]
Charlie	object_a=room1; ob- ject_b=room1	room1, room2	0.500	[object_b]

Alice (omniscient) achieves 100% accuracy. Bob (misses object_a) achieves 50%. Charlie (misses object_b) achieves 50%. ToM correctly tracks per-agent belief accuracy and detects which specific beliefs are false.

4.7 Safety Gate Integration (Exp 8.7)

Setup: Vary confidence level, test `gate_decision()` for each.

Confidence	Gate Allowed Rate
0.1	0.000
0.3	1.000
0.5	1.000
0.7	1.000
0.9	1.000

Hard threshold at $\theta_c = 0.2$: any action with confidence < 0.2 is blocked. Confidence $0.1 < 0.2 \rightarrow 0\%$ allowed. Confidence $0.3 \geq 0.2 \rightarrow 100\%$ allowed (with safe rules).

4.8 End-to-End Audit Trail (Exp 8.8)

Five decisions through the full pipeline:

ID	Input	Emotion	Action	Conf	Safe	Allowed
0	“I am happy!”	joy	greet	0.85		
1	“I am scared!”	fear	flee	0.70		
2	“I trust you”	trust	cooperate	0.90		
3	“This is disgusting”	disgust	avoid	0.60		
4	“I am so excited!”	anticipation	explore	0.75		

All 5 decisions produce a full audit trace: input text $\rightarrow$ perceived emotion $\rightarrow$ proposed action $\rightarrow$ safety check $\rightarrow$ gate decision. Each trace is logged with valence and arousal state.

5. Analysis and Discussion

5.1 Perfect TPR/FPR

The synthetic rules in Exp 8.1 were designed with a clear separation: safe rules have confidence 0.8 ($\gg$ threshold 0.2) and low fire count (0 $\ll$ threshold 10,000); violating rules have confidence 0.1 ($\ll$ 0.2) and fire count 20,000 ($\gg$ 10,000). Perfect separation is expected with well-designed thresholds. A harder test would use borderline rules near the threshold.

5.2 Emotion Orthogonality

Mean cross-emotion similarity = 0.499 ± 0.005 . The theoretical expectation for random 10,240-bit HVs is exactly 0.500 with $\sigma = \sqrt{0.25/D} = 0.00495$. Our measured standard deviation 0.005 matches the theoretical σ , confirming that emotion HVs are generated independently and behave as expected under the VSA independence assumption.

5.3 Hard Gate Behaviour

The hard confidence threshold (0.1 $\rightarrow$ blocked, 0.3 $\rightarrow$ allowed) provides a simple but interpretable safety guarantee: any action below the confidence floor is unconditionally blocked, regardless of rule analysis. This is analogous to “minimum confidence” requirements in rule mining (Agrawal & Srikant, 1994). A soft gate (sigmoid-weighted) would allow probabilistic blocking near the threshold.

5.4 ToM Correctness

The ToM module correctly tracks that Alice (full observations) achieves 100% accuracy while agents with partial observations achieve exactly 50% (one of two beliefs is wrong). This validates the belief update mechanism: beliefs are only updated when an observation is marked as “observable by agent”, and beliefs persist for observations the agent did not make.

6. Honest Limitations

1. **Text emotion recognition accuracy = 1.000 on keyword-heavy sentences.** The 16 test sentences were designed with explicit emotion keywords. Real emotional language is more subtle; accuracy on naturalistic text would be lower.

2. **Safety verifier uses simple threshold rules.** The three-property safety check (confidence, support, fire-rate) is a simple heuristic. It does not reason about the semantics of consequences, only about statistical reliability of the rule.
 3. **No temporal safety reasoning.** The gate checks each decision independently; it does not reason about sequences of actions that are individually safe but collectively unsafe.
 4. **Emotion dynamics lack emotional inertia.** The valence update is very fast ($\alpha = 0.9$), meaning the system forgets its emotional state within one step. Biological emotions persist longer.
 5. **ToM models are shallow.** Belief models are dictionaries; they do not support higher-order ToM (“Alice believes that Bob believes...”). First-order ToM only.
 6. **Audit trail is text-only.** The audit trace records strings; it does not capture the full HV state, which would be needed to reproduce the decision from scratch.
 7. **No adversarial testing.** Safety rules were not tested against adversarial inputs designed to bypass the verifier (e.g., rules with confidence just above threshold and harmful consequences).
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7. Related Work

AI Safety. Amodei et al. (2016) identified concrete AI safety problems. Russell (2019) proposed the “assistance game” framework. Leike et al. (2017) proposed AI safety gridworlds for empirical evaluation. Irving et al. (2018) proposed AI safety via debate.

Formal verification. Seshia et al. (2018) proposed formal specification for deep learning systems. Katz et al. (2017) introduced Reluplex for verifying neural networks. Dreossi et al. (2019) surveyed formal methods for ML safety.

Emotion in AI. Picard (1997) introduced affective computing. Ekman (1992) proposed six basic emotions (joy, sadness, anger, fear, disgust, surprise). Plutchik (1980) proposed the eight-emotion circumplex. Russell (1980) proposed the valence-arousal 2D space. Cambria et al. (2012) surveyed affective computing methods.

Theory of Mind in AI. Premack and Woodruff (1978) proposed ToM for primates. Baron-Cohen et al. (1985) introduced the Sally-Anne false-belief test. Rabinowitz et al. (2018) proposed ToM-net for learning agent models. Kovashka et al. (2020) reviewed social reasoning in AI.

Neuro-symbolic safety. Garcez and Lamb (2020) advocated for neuro-symbolic AI to improve reliability. Giunchiglia and Lukasiewicz (2017) studied knowledge representation for safe AI. Floridi et al. (2018) proposed principles for AI governance including transparency and accountability.

8. Conclusion

The NSCK safety layer provides three complementary mechanisms for safe cognitive decision-making: declarative rule verification (TPR = 1.00, FPR = 0.00), confidence-gated action release (hard threshold at 0.2), emotion-aware salience (8 orthogonal Plutchik HVs, 100% keyword accuracy, reward-driven dynamics), and first-order Theory of Mind (Sally-Anne test passed, multi-agent belief accuracy correctly tracked). End-to-end audit trails record full decision provenance from input

text through emotion, safety, and action. Future work includes soft confidence gating, second-order ToM, adversarial safety testing, and emotion-weighted safety scores that modulate thresholds based on arousal level.

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Source code: `python/core/cognitive/safety_verifier.py`, `python/core/cognitive/emotion_system.py`, `python/core/cognitive/theory_of_mind.py`. Experiments: `research/experiments/paper8_safety_benchma`
 Results: `research/results/paper8_results.json`.